



Week 1 Lecture

Unit Introduction

IFN647 -

Professor Yuefeng Li
School of Computer Science
Queensland University of Technology

Learning Objectives

Topic ID	Key topics	Mapping to slides	Activities
1	Overview	2-8	
2	Introduction of Nature Language Processing (NLP)	9-15	
3	Machine Learning Ethics and Example	16-20	
4	Information Retrieval (IR)	21-29	
5	NLP in the Era of LLMs	30-36	
6	Technologies used in IFN647: Python (.py and ipynb), Google, ChatGPT, Slack, Text processing in Python	37-41	Activities 1-4
7	Teaching Methodology	42-44	
	References	45	

1. Overview

- **This unit discusses** explores both the theory and practical applications of natural language processing (NLP) and machine learning (ML) techniques for text analysis.
- It discusses text processing techniques to represent text documents (unstructured), AI Ethical Issues and NLP in the Era of large language models (LLMs).
- It covers information retrieval (IR), Word Embedding, Generative Models, Indexing, Language models; ML algorithms (supervised, unsupervised and semi-supervised), User and Relevance models for
 - Information filtering, learning to rank,
 - Text classification, sentiment analysis,
 - Clustering, recommender systems,
 - Web search and topic modelling

IFN647 Overview

 chatgpt

×









AI Mode

All

Shopping

Images

News

Videos

Short videos

More

Tools

Sponsored results

 Chatbot App

https://chat.chatbot.app › chatgpt

Chatgpt 5.0 now available

GPT 5.0 now available — Meet the Chatgpt app , you can ask it any question and get a response in real time. Ask AI Anything — Ask AI to write anything and get instant answers. Chat with GPT 4.0. Try GPT 4o. Use GPT 4.

Chat with GPT

AI Chatbot developed on GPT Smart AI Chatbot : Chatbot App

GPT 4.0 now available.

Developed on GPT 4 Built on GPT 4o

Ask AI Anything - Chatgpt

Get AI Assistance for Homework, Day-To-Day Tasks, Essays, Content, Ideas, Translations etc

 ChatGPT

https://www.chatgpt.com

ChatGPT is available now

ChatGPT: AI at your fingertips — Discover a world of possibilities with ChatGPT. Easy to use, available anytime, anywhere. Try the...

ChatGPT

Free offer

PageRank was originally developed by Google?

Ask ChatGPT

Yes. PageRank was originally developed by Google—specifically by Larry Page and Sergey Brin at Stanford University, and it became a foundational algorithm behind early Google Search.













What traditional NLP techniques are needed for RAG-based prompt generation?

For RAG-based prompt generation, large language models get most of the attention—but traditional NLP techniques are still essential. They do the “quiet, structural work” that makes retrieval accurate and prompts reliable.

Here’s a clean breakdown, from data prep → retrieval → prompt construction.

Big Picture Summary

Traditional NLP in RAG mainly handles:

structure, precision, and control while LLMs handle generation and reasoning	
Stage	Traditional NLP Role
Indexing	Cleaning, chunking, metadata
Retrieval	BM25, keyword/entity matching
Ranking	Scoring, filtering
Prompting	Rule-based structure

1. Text Preprocessing & Normalisation

- Classic NLP, still unavoidable.
- Tokenisation (word / subword / sentence)
 - Lowercasing, punctuation handling
 - Stop-word removal (optional, task-dependent)
 - Stemming / Lemmatization
 - Sentence segmentation

Why it matters in RAG

- Improves chunking quality
- Reduces noise in sparse retrieval (e.g. BM25)
- Helps build consistent metadata for indexing

2. Document Chunking & Segmentation

Mostly rule-based NLP + heuristics.

- Fixed-length chunking (tokens / sentences)
- Discourse-aware chunking (paragraphs, sections)
- Overlapping windows
- Heading-based segmentation

Why

- Retrieval operates at chunk level, not document level
- Bad chunking = good info never retrieved
- Impacts context window efficiency

3. Sparse Retrieval Techniques (Classical IR)

Still very strong, especially for exact matches.

- TF-IDF
- BM25 / BM25+
- Inverted indexes
- Query expansion (synonyms, keywords)

Related Knowledge

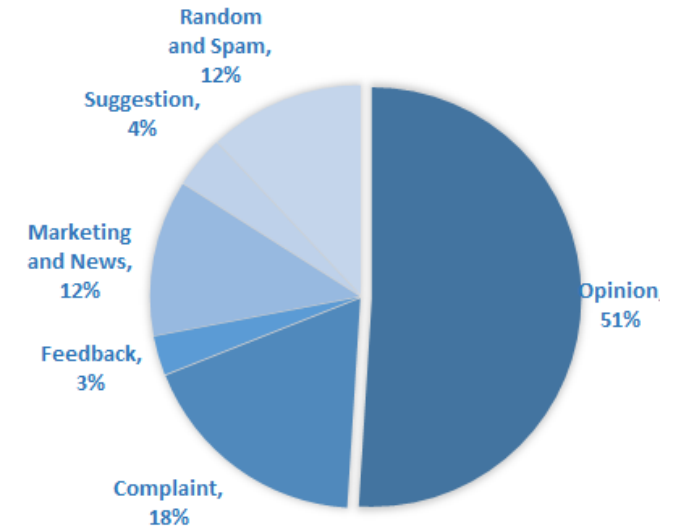
- **Pre-requisite:**
 - (IFN509 or IFQ509 or IFN580) OR (192cp in IV04) OR (admission into IV54)
- **Assumed Knowledge**
 - Programming languages experience (e.g., IFN555)
 - Basic statistics functions and data mining (required by IFN509)
 - Machine learning (Required by IFN580)

Weekly Schedule and Assessment Items

- You can find the weekly schedule in **Course Syllabus** of the IFN647 Canvas
- Assessment items can be found in **Assignments** of the IFN647 Canvas
 - Assignment 1 (individual, 20%)
 - A portfolio of work completed from week 2 to week 7
 - Assignment 2 (in group of 3 people, 35%)
 - A major text analysis project which may be undertaken in a team of 3 people.
 - Deliverables:
 - In-Lab or Online marking of Project Demo – 5% component, week 13 workshop time.
 - Final Report – 30% component, week 14, submit via Canvas.
 - Final Exam (individual, 45%)
 - A written examination (**an in person invigilated exam**), which covers the material presented in the lectures and workshops throughout the semester.

Why this unit is important?

- Big data is a collection of data so large and complex that it becomes very difficult to process. It extends beyond structured data, including unstructured data: text, audio, video, click streams, log files and more.
- It can contribute to the velocity of big data by facilitating rapid decisions that lead to other decisions. E.g., the world's stock trades, or Web search engines are done using AI-based systems.
- It can be used to mitigate the problem of variety by capturing, structuring, and understanding unstructured data or information. E.g., understanding the impact of unstructured data (text) issues such as a firm's reputation, or intent analysis of comments.
- AI can be used to identify and clean dirty data (an issue for volume) or use dirty data as a means of establishing context knowledge for the data.
- It can make intelligent data visualization apps available, possibly for particular types of data.
- AI can be used to process and interpret knowledge or insights in natural languages and/or other forms of data , e.g., audio, video or image data.



Top 5 S&P/ASX200					
Gains	Declines	Value	Volume		
		last price	change		
GXY	GALAXY RES	\$1.385	↑ 5.73%	\$0.075	
SDA	SPEEDCAST	\$1.965	↑ 3.42%	\$0.065	
HUB	HUB24LTD	\$11.64	↑ 3.01%	\$0.34	
RSG	RESOLUTE	\$1.565	↑ 2.96%	\$0.045	
NWH	NRWHOLDLTD	\$2.345	↑ 2.85%	\$0.065	

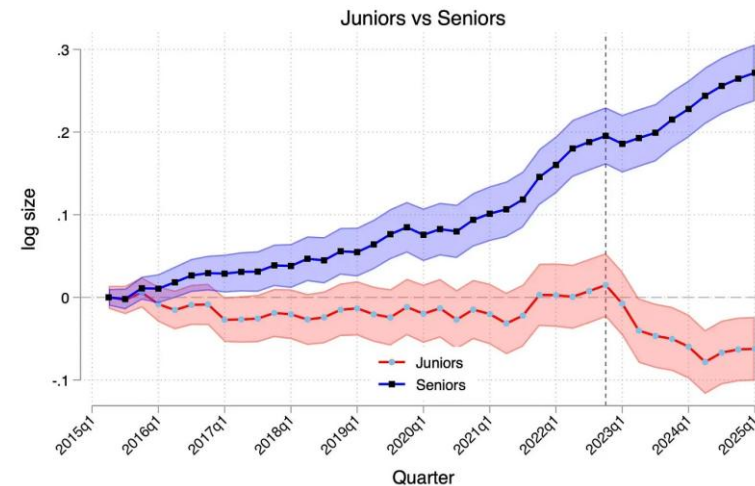
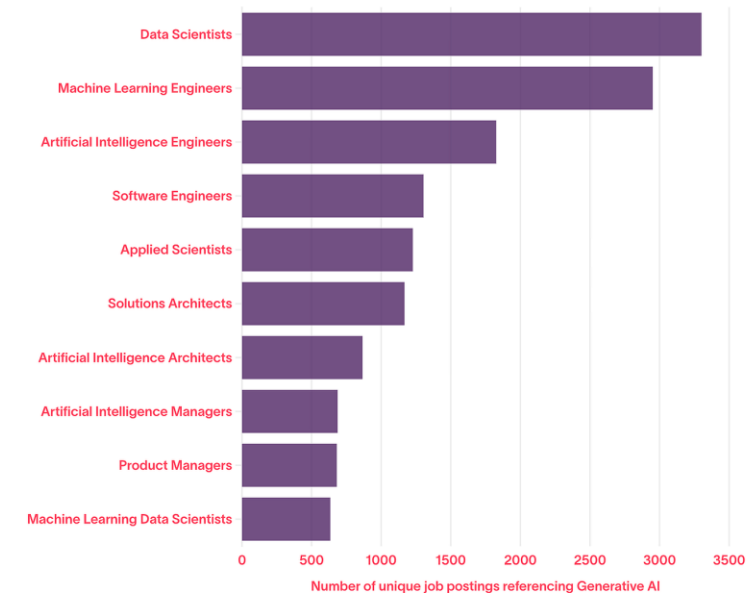
Why this unit is important? Cont.

- It is estimated that more than 80% of big data were stored in text format, which is creating a number of issues and challenges for NLP such as the noisy and uncertain information, scalability and effectiveness.
- Big data is more than a challenge for enterprisers. It is also an opportunity to find insight in new and emerging types of data, to make business more agile, and to answer questions that, in the past, were beyond reach.
- Therefore, there is a significant need for more efficient and effective techniques to facilitate the process.
- [Australia's Top 10 In Demand Jobs in AI & Data for 2026](#) (Google)
 - AI/Machine Learning Engineer (Australian companies are increasingly leveraging AI to drive innovation and efficiency)
 - Data Scientist
 - Data Engineer
 - LLM/Generative AI Engineer
- [AI and Data Scientist Job Market in 2026](#) (Google)
 - The need for data scientists is not disappearing; in fact, there is an increasing demand for data talent that knows AI
 - Senior roles are benefiting the most from AI adoption

© COS [National AI Plan](#)

Top job titles for postings referencing Generative AI

May 2024 - May 2025



2. Introduction of Nature Language Processing (NLP)

- **Natural Language Processing (NLP)** is a subfield of Artificial Intelligence (**AI**) that uses Machine Learning (**ML**) techniques to enable machines to effectively analyze and understand human language.
- It integrates methods from linguistics, computer science, and machine learning to represent unstructured text data in machine-readable, structured formats, bridging the gap between human communication and machine processing, such as understanding, interpreting, and generating language.
- NLP can also extract information from large collections of textual documents, typically by discovering topics, patterns, and trends using learning algorithms.
- It powers many widely used applications, including speech recognition, chatbots, language translation, sentiment analysis, search engines, and document summarization across numerous industry sectors.
- Modern NLP increasingly relies on deep learning and large language models (**LLMs**), enabling systems to learn rich patterns from vast amounts of text and perform complex language tasks with high accuracy.
 - In the era of LLMs, there is a significant paradigm shift for NLP research from task-specific models (e.g., classification, named entity recognition) to general-purpose foundation models pretrained on massive corpora.
 - Also, a single LLM can perform multiple NLP tasks (QA, summarization, translation, explanation, retrieval or filtering) via prompting or fine-tuning.
 - Contextual, deep semantic representations replace handcrafted features and shallow embeddings (e.g., BERT).
 - Learning with Less Supervision (e.g., Zero-shot, few-shot, and instruction-based prompting).

CRICOS No.00213J

Text Data Engineering

- **Text data engineering** is a specialized area within data engineering that focuses on building systems to enable the collection, processing, and utilisation of text data, which may exist in a variety of formats.
- It involves designing pipelines to collect, clean, transform, and store raw, unstructured text so that it becomes reliable, high-quality input for AI-based applications such as Large Language Models (LLMs).
- Text data can appear in several formats:
 - **Unstructured data:** has no predefined structure or data model (e.g., emails with free-text content, HTML pages).
 - **Semi-structured data:** contains self-describing structure but does not follow rigid schemas (e.g., XML, JSON, delimiter-separated values files).
 - **Structured data:** stored in well-defined formats and repositories (e.g., relational SQL databases).
- Among these, four of the most common formats used to mark up or structure text data are **HTML**, **XML**, **JSON**, and **CSV**.

CRICOS No.00213J

Examples of Text Formats: Free-text and HTML

A Tweet:

Donald J. Trump Verified account @realDonaldTrump

The NFL National Anthem Debate is alive and well again - can't believe it! Isn't it in contract that players must stand at attention, hand on heart? The \$40,000,000 Commissioner must now make a stand. First time kneeling, out for game. Second time kneeling, out for season/no pay!

A HTML file:

```
<!DOCTYPE html>
```

```
<html lang="en" data-scribe-reduced-action-queue="true">
```

```
<head>
```

```
<link rel="search" type="application/opensearchdescription+xml" href="/opensearch.xml" title="Twitter">
```

```
<meta property="og:title" content="Donald J. Trump on Twitter">
```

```
<meta property="og:description" content="The NFL National Anthem Debate is alive and well again - can't believe it! Isn't it in contract that players must stand at attention, hand on heart? The $40,000,000 Commissioner must now make a stand. First time kneeling, out for game. Second time kneeling, out for season/no pay!">
```

```
<meta property="og:site_name" content="Twitter">
```

```
</head>
```

Examples of Text Formats: XML example

12

```
<newsitem itemid="7622" id="root" date="1996-08-22" xml:lang="en">
<title>UK: British economy grew 0.4 percent in second quarter.</title>
<headline>British economy grew 0.4 percent in second quarter.</headline>
<dateline>LONDON 1996-08-22</dateline>
<text>
<p>The British economy grew by 0.4 percent in the second quarter of 1996, and by 1.8 percent compared with the second quarter of 1995, the Office for National Statistics said on Thursday. </p>
<p>The figures were unchanged from the previous official estimate.</p>
<p>The Treasury, commenting on the figures, said it expected growth to be stronger in the second half of the year.</p>
</text>
<copyright>(c) Reuters Limited 1996</copyright>
<metadata>
<codes class="bip:countries:1.0">
<code code="UK">
<editdetail attribution="Reuters BIP Coding Group" action="confirmed" date="1996-08-22"/>
</code>
</metadata>
```

Examples of Text Formats: JSON files

13

- JSON (Javascript Object Notation) is a **data interchange format**.
- Readable to humans & can be transmitted/interpreted by computer.
- Data represented as **attribute-value pairs**.
- JSON can have **nested structure**. XML, HTML also have nested data.
- **CLI** tools treat files as **plain text** (i.e. no explicit structure, or only contain very simple delimiter-separated rows) CLI commands seen so far may be difficult to apply to JSON. Two solutions:
 - use command line tools designed to operate on such data
 - converted nested/structured data into tabular plain text
- Read .json files in Python

CRICOS No.00213J

```
{
  "first_name": "John",
  "last_name": "Smith",
  "is_alive": true,
  "age": 27,
  "address": {
    "street_address": "21 2nd Street",
    "city": "New York",
    "state": "NY",
    "postal_code": "10021-3100"
  },
  "phone_numbers": [
    {
      "type": "home",
      "number": "212 555-1234"
    },
    {
      "type": "office",
      "number": "646 555-4567"
    }
  ],
  "children": [
    "Catherine",
    "Thomas",
    "Trevor"
  ],
  "spouse": null
}
```

Attributes

Values

Values can be:

- strings (*John*)
- digits (*25*)
- objects (like for *address*)
- arrays (like for *children*)
- boolean (*true*)
- the *null* value (as for *spouse*)

Examples of Text Formats: Delimiter-separated files

- They are used to store data expressed in two-dimensional arrays $A(i, j)$

Student ID	First name	Last name	Grade
3050395	John	Smith-Jones	6
4749573	Jessica Anna	Johnson	5
3729173	Marie	Green	7

- Rows can be separated by new line character ($\backslash n$)
- Columns can be separated using a delimiter character
 - commas (CSV), space, tabs (TSV)
- Please note column headers are sometimes included in the first row of the file.

Documents vs. Database Records

- Database records (or *tuples* in relational databases) are typically made up of well-defined fields (or *attributes*)
 - E.g., bank records with account numbers, balances, names, addresses, social security numbers, dates of birth, etc.
 - They are **structured**
 - Easy to compare fields with well-defined semantics to queries to find matches
- Text is more difficult !!!
 - Text documents are **unstructured or semi-structured**
- Example bank database query
 - *Find records with balance > \$50,000 in branches located in Amherst, MA.*
 - Matches easily found by comparison with field values of records by using a SQL query:
SELECT * FROM *accounts*
WHERE balance > 50000 AND branch_city = 'Amherst' AND branch_state = 'MA';
- Example search engine query
 - *“bank scandals in western mass”*
 - This text must be compared to the text of entire news stories, and consider semantic issues, e.g.,
 - Synonym of “scandal”: *Disgrace Dishonor or Indignity*;
 - Stemming (e.g., plural): *Scandals, Disgraces, Dishonors, Indignities*

3. Machine Learning and ML Ethics



- Machine Learning (ML), a core subfield of Artificial Intelligence (AI), enables systems to automatically learn models or functions directly from data, rather than relying on manually engineered rules. This capability makes ML a scalable and flexible alternative to traditional rule-based systems.
- Deep Learning, a specialized area within ML, utilizes multi-layer neural networks to learn hierarchical representations from large-scale data and perform complex tasks. It has driven substantial improvements in state-of-the-art performance across domains such as computer vision, speech recognition, natural language processing (NLP), recommender systems, and many others.
- Machine learning is widely regarded as a key driver of the next major wave of technological innovation.

Machine Learning Risks

- As recognized in the Bletchley Declaration of the 2023 AI Security Summit, while AI offers substantial opportunities, it also introduces significant risks.
- Key risk categories include [1]:
 - **Malicious use:** Adversarial actors may exploit AI systems to cause large-scale harm. For example, AI could centralize control over trusted information if only a small number of actors possess the technical expertise and resources required to develop cutting-edge models, enabling them to propagate preferred narratives at scale.
 - **AI race dynamics:** Competitive pressures may incentivize the deployment of AI systems in unsafe or insufficiently tested ways, even when such actions are contrary to collective interests. AI agents may operate autonomously or exhibit self-interested behaviors, becoming increasingly capable yet less interpretable to humans.
 - **Organizational risks:** Accidents may arise from both the inherent complexity of advanced AI systems and the organizations that develop and deploy them. In deep learning–based systems, limited transparency makes it difficult to predict system behavior or understand how far capabilities may exceed original expectations.
 - **Rogue AI risks:** Controlling systems that surpass human intelligence poses fundamental challenges. AI agents are often goal-driven (e.g., exhibiting proxy gaming behaviors), raising the risk that humans may lose effective control over the objectives that such systems ultimately pursue.

CRICOS No.00213J

Machine Learning Ethics

- It also noted that the ability to manipulate or generate content by using Generative AI (GenAI) may pose unforeseen risks, or a variety of novel ethical, legal, social, and technological challenges. Concerns have also arisen about **ethics, privacy** and security in AI.
- **Machine learning ethics refers** to the study of the moral principles involved in designing, implementing, and deploying machine learning algorithms.
 - It assesses how these algorithms can impact individuals, society, and various industries. The objective of machine learning ethics is to establish guidelines that promote **transparency, accountability, accuracy and fairness**.
- Higher transparency is a common dimension when discussing ethics of ML algorithms.
- To maximize the benefits of AI or ML while at the same time mitigating or even preventing its risks and dangers, the concept of trustworthy AI (TAI) has been proposed if trust can be established in its development, deployment, and use;
 - As AI-based autonomous systems are generative and learn, evolve and permanently change their functional capacities.

Machine Learning Ethics cont.

19

- Explicability and bias are two central concepts within ML ethics and both directly connect to ethical dimensions: fairness, accuracy, accountability, and transparency
 - **Explicability** is a TAI principle for achieving higher transparency, which comprises an epistemological sense as well as an ethical sense. In its epistemological sense, explicability entails the creation of explainable AI by producing (more) interpretable AI models whilst maintaining high levels of performance and accuracy.
 - In its ethical sense, explicability comprises the creation of accountable AI or ML.
 - For example, if a ML model rejects a loan application, the applicant deserves to know why.
 - **Bias** may lead to an unfair ML algorithm, which occurs when two data sets are not considered equal, possibly due to biased assumptions in the AI algorithm development process or built-in prejudices in the training data.
 - So, it is important to identify how bias can creep in and put in place appropriate internal controls to address the concerns.
 - Examples: A ML model may be accurate overall but inaccurate for specific subgroups. If biased outcomes cause harm (e.g., medical misdiagnosis), someone must be responsible.
- **Ethical issues** are situations where ethical decisions need to be made, for example, what is the precision threshold and data quality indicator when validating ML algorithms; how to protect the privacy of users when deploying ML algorithms (whether encryption, encoding, or federated ML meets the requirements).

CRICOS No.00213J

Example 1: Sentiment Analysis (opinion mining)

- Blogs, online reviews, and forum posts are often opinionated, e.g.,
 - “The **TV**’s sound is *good*, but the resolution quality looks *mediocre*.”
 - Where **TV** is an entity, sound & resolution are aspects or features of TV and *good* or *mediocre* are opinion words.
- Sentiment classification attempts to automatically identify the polarity of the opinion for a given review.
 - Negative opinion
 - Neutral opinion
 - Positive opinion
- Sometimes the strength of the opinion is also important
 - “Two stars ** ” vs. “four stars **** ”
 - Weakly negative vs. strongly negative
- **Ethical Issue:** How is the personal information of reviewers used?
 - Read National AI Ethics Principles, e.g.,
 - [Australia’s AI Ethics Principles | Department of Industry Science and Resources](#)

All user reviews

General Comments (148 comments)



Ease of Use (108 comments)



Screen (92 comments)



Software (78 comments)



Sound Quality (59 comments)



Size (59 comments)



4. Information Retrieval (IR)

- It is the process of searching and presenting relevant information from a collection of data or documents to meet a user's needs (or a query).
- IR is more than just text, and more than just web search
 - although these are central
- People doing IR work with different media (e.g., video, photos, music, speech), different types of search applications, and different tasks
- Comparing Text
 - Comparing the **query text** to the **document text** and determining what is a **good match** is the core issue of IR
 - Exact matching of words is not enough
 - Many different ways to write the same thing in a “natural language” like English
 - e.g., does a news story containing the text “*bank director in Amherst steals funds*” match the query?
 - Some stories will be better matches than others

Dimensions of IR

Content	Applications	Tasks
Text	Web search	Ad hoc search
Images	Vertical search	Filtering
Video	Enterprise search	Classification
Scanned docs	Desktop search	Question answering
Audio	Forum search	
Music	P2P search	
	Literature search	

Big Issues in IR: relevance

- What is it?
- Simple (and simplistic) definition: A relevant document contains the information that a person was looking for when they submitted a query to the search engine
- Many factors influence a person's decision about what is relevant: e.g., task, context, novelty, style
- *Topical relevance* (same topics) vs. *user relevance* (everything else)

Query = “australia great barrier reef”

All
 Images
 News
 Videos
 Maps
 More

Tools

About 25,800,000 results (0.52 seconds)

Ad · <https://www.viator.com/tours-&-tickets/book-now>

Great Reef Barrier - Book Tours & Tickets on Viator

Quick & Easy Purchase Process! Full Refund Available up to 24 Hours Before Your Tour Date. **Great Barrier Reef** Tours & Tickets are Selling Out Fast. Book Now to Avoid Disappointment. Low Price Guarantee. Skip the Line Tickets. Photos,...

★★★★★ Rating for viator.com: 4.3 - 2,975 reviews

Secure Checkout
We Use SSL Encryption to Keep Your Data Safe & Secure

Reserve Now & Pay Later
Keep Your Plans Flexible. Reserve Your Activity & Pay Later

Ad · <https://www.getyourguide.com/getyourguide/activities>

Australia Great Barrier Reef - Tours, Tickets and Activities

Make the most of each moment when you travel. Skip the hassle and book in advance. Convenient cancellation up to 24 hours before your event — no...

[Lady Musgrave Experience](#) · [Glass Bottom Boat Cairns](#) · [Snorkelling Tours](#)

Ad · <https://www.redballoon.com.au/>

Great Barrier Reef Tours, Snorkelling & Cruises | RedBalloon

Buy Or Gift From RedBalloon With Thousands Of Experiences **Australia**-Wide. Support Local. AfterPay and Zip Available. Vouchers Valid for 5 Years And Emailed Instantly. Shop...

<https://greatbarrierreef.org>

Great Barrier Reef | Australia's Great Natural Wonder

One of **Australia's** most remarkable natural gifts, the **Great Barrier Reef** is blessed with the breathtaking beauty of the world's largest coral reef.

[About The Reef](#) · [Facts](#) · [The Reef From The Air](#) · [Sail the Reef](#)

Great Barrier Reef

Coral reef in Australia

The Great Barrier Reef is the world's largest coral reef system composed of over 2,900 individual reefs and 900 islands stretching for over 2,300 kilometres over an area of approximately 344,400 square kilometres. [Wikipedia](#)

Area: 348,700 km²

Location: Off the east coast of the [Queensland](#) mainland, [Australia](#)

UNESCO Site Id: 154

Criteria: Natural: vii, viii, ix, x

Importance

Function

Age

Weather

Animals [Great Barrier Reef](#) View 45+ more

Example of Relevance

Query = “澳大利亚大堡礁”

澳大利亚大堡礁

✕

🔊

📷

🔍

🔍 All

📍 Maps

🖼️ Images

📰 News

📺 Videos

⋮ More

Tools

About 630,000 results (0.37 seconds)

https://www.australia.cn › places › gu... · Translate this page

大堡礁指南 - 澳大利亚旅游局

大堡礁纵贯澳大利亚东部海岸线，绵延2,300 多公里，带给你丰富多彩的沿海体验，这是
在其他任何地方都无法想象的。这里有令人惊叹的珊瑚丛、巨型蛤蜊、稀有的鲸鱼以及 ...

https://baike.baidu.com › item › 大堡... · Translate this page

大堡礁（世界自然遗产-位于澳大利亚东岸昆士兰州的大 ...

大堡礁位于澳大利亚东北海岸，这里物种多样、景色迷人，有着世界上最大的珊瑚礁群，
包括400种珊瑚、1500种鱼类和4000种软体动物。大堡礁还是一处得天独厚的科学研究场
所， ...

地理位置: 澳大利亚昆士兰州 外文名: Great Barrier Reef
开放时间: 全年 景点级别: 世界自然遗产

遗产描述 · 遗产介绍 · 百态珊瑚 · 环境保护

https://prodivercairns.com › thereof · Translate this page

大堡礁| Pro Dive Cairns | 澳大利亚

澳大利亚大堡礁为世界七大自然奇观及联合国教育科学文化组织指定世界自然遗产。以其
独特的生物多样性和海洋生物而闻名，如珊瑚，乌龟，鲨鱼，巨蛤和各种鱼类。

https://www.australia.com › places · Translate this page

大堡礁指南 澳洲旅遊局

大堡礁在澳洲東岸水域延綿超過2,300公里（1,430英里），提供的海岸體驗比全球任何地方更豐
富。在迷人的珊瑚礁、巨型蛤蚌和珍罕的鯨魚類之中暢泳，並且世上七個海龜類中您可 ...

https://zh.wikipedia.org › zh-my › 大... · Translate this page

大堡礁- 维基百科，自由的百科全书 - Wikipedia

大堡礁（英语：Great Barrier Reef）是世界最大最长的珊瑚礁群，位于南太平洋的澳洲东北海
岸，它纵贯于澳洲东北昆士兰州外的珊瑚海，北从托雷斯海峡，南到南回归线以南（约 ...

Great Barrier Reef (大堡礁)

Coral reef in Australia

The Great Barrier Reef is the world's largest coral reef system composed of over 2,900 individual reefs and 900 islands stretching for over 2,300 kilometres over an area of approximately 344,400 square kilometres. [Wikipedia](#)

Area: 348,700 km²

Location: Off the east coast of the [Queensland](#) mainland, [Australia](#)

UNESCO Site Id: 154

Criteria: Natural: vii, viii, ix, x

Importance

Function

Age

Weather

Animals Great Barrier Reef

View 45+ more

CRICOS No.00213J

Professor Yuefeng Li
School of Computer Science

Example of Relevance cont.

Query = “澳大利亚大堡礁”

澳大利亚大堡礁 - 百度百科



简介：澳大利亚大堡礁一般指大堡礁。大堡礁（英文：The Great Barrier Reef，法文：Grande barrière de corail），是世界最大最长的珊瑚礁群，位于南半球，它纵贯于澳大利亚的东北沿海昆士兰州，北从托雷斯海峡，南到南回归线以南，绵延伸展共有2011公里，...

基本资料 遗产描述 机构评价 遗产介绍 主要景点 更多>

百度百科

澳大利亚大堡礁 - 百度图片



image.baidu.com 查看全部15,560 张图片

2022年澳大利亚的大堡礁,海量收录.完整版

相关地学术语

展开



航拍绿岛大堡礁



贝里斯堡礁



日积礁



世界十大岛屿



托雷斯海峡群岛



巴西大西洋群岛

百度热搜

换一换

权威解读当前经济热点问题

1 多地宣布无症状或轻症可上班 热

2 专家：北京未来一两周将迎重症高峰 热

3 我阳了，我好了

4 周围人全阳了 他们为何至今未感染? 热

5 国家要求快递业1月8日停业?真相...

6 两架飞机在韩国仁川机场相撞

7 阿根廷门将领奖台不做不雅动作惹争议

How to use relevance?

- Information retrieval (IR) models define a view of relevance.
- *Ranking algorithms* used in search engines are based on IR models.
- Most traditional IR models describe **statistical properties** of text rather than linguistic
 - Statistical approach to text processing started with Luhn in the 50s
 - E.g., counting (term frequency) simple text features such as words instead of parsing and analyzing the sentences.
 - Linguistic features (e.g., parts of speech (POS) tags) can be part of a statistical model.
 - There are eight parts of speech in the English language: **noun, pronoun, verb, adjective, adverb, preposition, conjunction, and interjection.**
- Machine learning algorithms can be used to classify relevant and non-relevant documents for a given query, where text data should be represented as numerical data vectors or matrix.

Big Issues in IR: evaluation

- Experimental procedures and measures for comparing system output with user expectations
 - Originated in Cranfield experiments in the 60s
- IR evaluation methods now used in many fields such as machine learning and data mining.
- Typically use *test collection* of documents, queries (or topics), and relevance judgments
 - Most commonly used are TREC (Text REtrieval Conference) collections
 - <https://trec.nist.gov/>
- *Recall* and *precision* are two examples of effectiveness measures.

Big Issues in IR: Users and Information Needs

- Search and classification evaluation is user-centered.
- Keyword queries are often poor descriptions of actual information needs.
- Interaction and context are important for understanding user intent.
- Query refinement techniques such as *query expansion*, *query suggestion*, *relevance feedback* and *training sample generation* improve ranking (we will discuss these techniques in this unit).

5. NLP in the Era of Large Language Models (LLMs)

- **Paradigm Shift:** From task-specific models (e.g., classification, tagging) to **general-purpose foundation models** pretrained on massive corpora.
- **Unified Modeling:** A single LLM can perform **multiple NLP tasks** (QA, summarization, translation, explanation, sentiment, IR) via prompting or fine-tuning.
- **Representation Learning:** Contextual, deep semantic representations replace handcrafted features and shallow embeddings (e.g., BERT).
- **Learning with Less Supervision:** Zero-shot, few-shot, and instruction-based learning reduce reliance on large labeled datasets.
- **Key Challenges:** Hallucination, bias, explainability, efficiency, data privacy, and evaluation reliability.

LMs, PLMs and LLMs

- Language Models (LMs)
 - Discuss probability of next word (we will discuss it in week 5). For example,
 - n-grams, discuss next word probability from counts:

$$p(w_2 \mid w_1) = \frac{\text{Count}("w_1w_2")}{\text{Count}("w_1")}$$

- neural LMs, discuss next word probability from neural net

$$p(w_i \mid w_{i-2}, w_{i-1})$$

- Pre-trained Language Model (PLM) –
 - E.g., BERT, one advanced word embedding technique
- Large Language Model (LLM) – GPT-3.5
 - E.g., various prompting approaches

Prompting Methods for LLMs

- **Zero-shot Prompting**

Model performs a task using only a natural language instruction, without examples.

- **Few-shot Prompting**

Provide a small number of input–output examples in the prompt to guide the model.

- **Instruction Prompting**

Explicitly describe the task, constraints, and expected output format.

- **Chain-of-Thought (CoT) Prompting**

Encourage step-by-step reasoning to improve performance on reasoning-intensive tasks.

- **An example prompt:**

Translate English to Chinese.

[en]: Structured prompting enforces specific output formats. [ch]: 结构化提示强制执行特定的输出格式.

[en]: RAG can inject external knowledge into the prompts.

[ch]: RAG可以将外部知识融入提示中.

[en]: Role prompting can assign a model a role (e.g., “You are an information extraction system”) to shape style and reasoning.

[ch]:

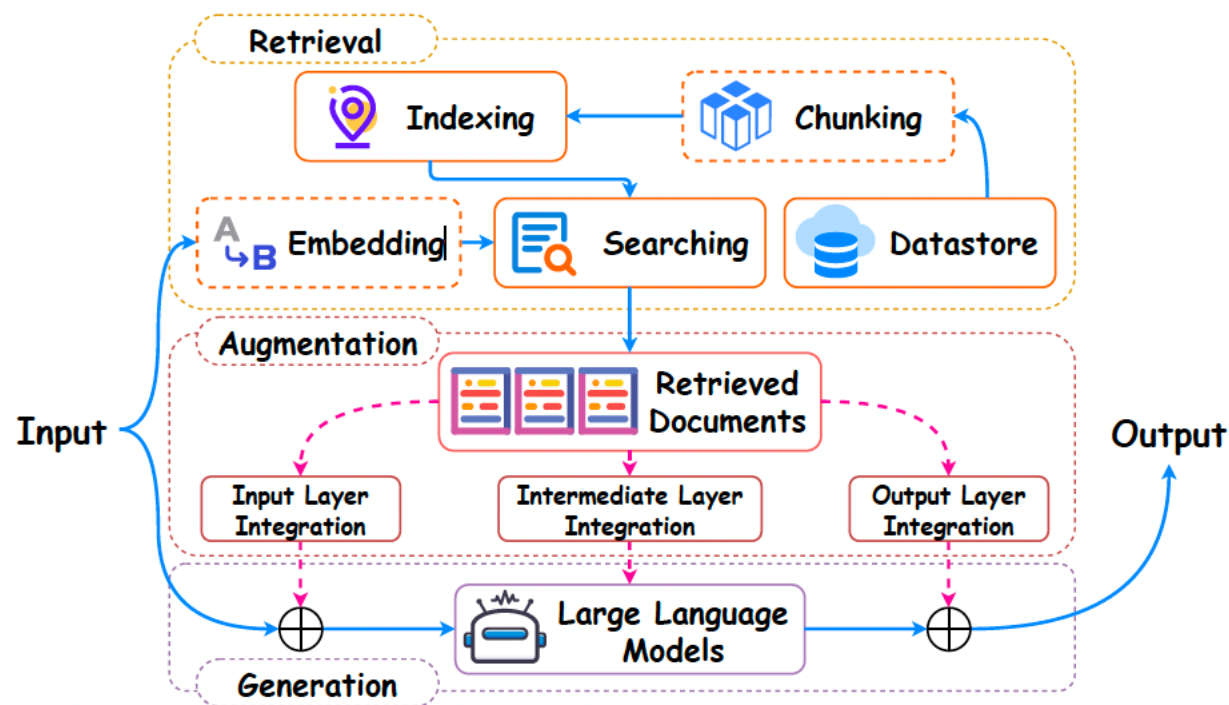
- askChaGPT

[en]: Role prompting can assign a model a role (e.g., “You are an information extraction system”) to shape style and reasoning.

[ch]: 角色提示可以为模型指定一个角色（例如“你是一个信息抽取系统”），以引导其输出风格和推理方式。

Retrieval Augmented Generation (RAG)

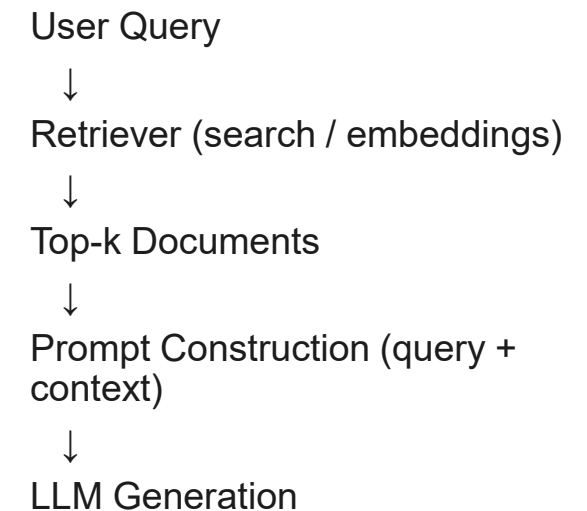
- Figure on the right [3] shows an illustration of the basic RAG-based LLMs framework, which consists of three major components: retrieval, generation, and augmentation.
 - Retrieval includes different procedures depending on the specific designs. The retrieved documents are further leveraged in generation with the augmentation module, which may be at different integration stages.



Python Example of RAG-based Prompt Generation

- **Assume:**
 - `query` = "What are the limitations of large language models?"
 - `retrieved_docs` = [
 "Large language models may hallucinate facts when lacking grounding.",
 "LLMs require significant computational resources for training.",
 "Bias in training data can affect model outputs."
]
 - `prompt` = `build_rag_prompt(query, retrieved_docs)`
- In the era of LLMs, this unit equips you with the skills to search `retrieved_docs` within a document set for a given query and to design and develop various forms of the `build_rag_prompt()` function, enabling the generation of precise and informative responses to **user information needs**.
- It also prepares you to perform related tasks for different information needs, such as sentiment analysis, news recommendation, and more.

- **Basic RAG Procedure:**



Rethinking Techniques in the Era of LLMs

- LLMs have significantly lowered the barrier to performing coding and NLP tasks: explicit model training is often unnecessary, as tasks can be addressed by interacting with an LLM API or web interface (e.g., ChatGPT).
- However, the effectiveness of these models is highly dependent on **prompt design**. Well-structured prompts are critical for eliciting accurate, coherent, and contextually relevant outputs.
- One promising approach to enhance prompt quality is **RAG** (as illustrated in the procedure on the right). This process typically involves several steps:
 - **Information Retrieval (IR)**: Identify candidate documents relevant to the user query from external knowledge or resources.
 - **Document Representation and Filtering**: Encode documents using word embeddings and apply **information filtering** to select the top-k candidates. Techniques such as **learning-to-rank**, **classification**, or **clustering** can be used to prioritize highly relevant documents.
 - **Prompt Construction**: Integrate query and context into the prompt, potentially augmented by:
 - **Query Expansion, Information Extraction, Sentiment or Context Analysis**: Ensure that prompts reflect the intended tone or focus.
 - **Language Modeling**: Provide contextual grounding for the LLM, enabling generation of precise and informative responses.
- This workflow highlights that **traditional NLP techniques remain indispensable** in the era of LLMs.

Overview of LLM-Related Topics for This Unit

- Bidirectional Encoder Representations from Transformers (BERT)
- LLM-based Information Extraction
- LLM-based Information Filtering
- Chain-of-Thought (CoT) Prompting for Query Expansion
- Issues in LLM-based social search
- Sentiment Analysis in the Era of LLMs: text classification, aspect-based sentiment analysis and Analysis of Subjective Texts
- LLM-Era clustering
- LLMs for News Recommendation

6. Technologies Used in IFN647

- Python 3.13.5, PyCharm (2025) and Anaconda (Jupyter Notebook, 2025).
 - Basic Python proficiency, including common commands used in both .py and .ipynb files.
 - Familiarity with text processing in Python.
 - Working knowledge of some common NLP libraries, e.g., html.parser module, and packages: nltk, gensim, pandas, torch, transformers, etc.
- Ability to use Google and ChatGPT to demonstrate IR, web search, and LLM-based prompting approaches.
- Use of Slack and email for communication.
- **Example** text data – an XML document

```
<?xml version="1.0" encoding="ISO-8859-1"?>
- <newsitem xml:lang="en" date="1997-08-06" id="root" itemid="783802">
  <title>UK: Britain's Channel 5 to broadcast Fashion Awards.</title>
  <headline>Britain's Channel 5 to broadcast Fashion Awards.</headline>
  <dateline>LONDON 1997-08-06</dateline>
  - <text>
    <p>Channel 5, Britain's newest terrestrial television channel, on Wednesday said it struck a three-year deal with the British Fashion Council for exclusive rights to broadcast the Lloyds Bank British Fashion Awards.</p>
    <p>"Fashion is now a defining part of British popular culture and this was an unmissable opportunity for 5 to give the British Fashion Awards its rightful place and profile on television," Adam Perry, Channel 5's controller of events, said in a statement.</p>
    <p>Initial, the independent production company that will produce coverage of the awards, said it aims to bring in top celebrities and world-class performers for the event.</p>
    <p>The British Fashion Awards will be held on October 22 at London's Royal Albert Hall.</p>
    <p>-- London Advertising Newsdesk +44 171 542 2815</p>
  </text>
  <copyright>(c) Reuters Limited 1997</copyright>
</newsitem>
```

Python Introduction:

- Python is an easy-to-learn yet powerful programming language for text processing.
- It provides efficient high-level data structures and a simple but effective approach to object-oriented programming.
- Python's elegant syntax and dynamic typing, together with its interpreted nature, make it well suited for scripting and rapid application development across many domains and platforms.
- Python introduction tutorial: self-reading this week to ensure familiarity with commonly used commands.
 - Python overview
 - Program and functions
 - Dictionaries, Structuring Data and Classes
 - Access Files and the Web

Example of text processing in Python

- A task for the example XML document:
 - Print out the news itemid, the number of words in the news, and the news content (in lines).

```
import string
myfile=open('783802.xml', 'r')
start_end = False
file_=myfile.readlines()
word_count = 0
news=""
for line in file_:
    line = line.strip()
    if(start_end == False):
        if line.startswith("<newsitem "):
            for part in line.split():
                if part.startswith("itemid="):
                    docid = part.split("=")[1].split("\n")[1]
                    break
            if line.startswith("<text>"):
                start_end = True
        elif line.startswith("</text>"):
            break
        else:
            line = line.replace("<p>", "").replace("</p>", "")
            news = news + line + "\n"
            for term in line.split():
                word_count += 1
myfile.close()
print('The news document itemid: '+ docid+ ' contains: '+ str(word_count) + ' words')
print("-----")
print(news)
```


Find lines between the line `<text>` and the line `</text>`.

else:

from here `start_end = True`

PS: The difference between different editors of XML documents; e.g., `'"'`

```
<?xml version="1.0" encoding="ISO-8859-1"?>
- <newsitem xml:lang="en" date="1997-08-06" id="root" itemid="783802">
  <title>UK: Britain's Channel 5 to broadcast Fashion Awards.</title>
  <headline>Britain's Channel 5 to broadcast Fashion Awards.</headline>
  <dateline>LONDON 1997-08-06</dateline>
  - <text>
    <p>Channel 5, Britain's newest terrestrial television channel, on Wednesday said it struck a
      three-year deal with the British Fashion Council for exclusive rights to broadcast the
      Lloyds Bank British Fashion Awards.</p>
    <p>"Fashion is now a defining part of British popular culture and this was an unmissable
      opportunity for 5 to give the British Fashion Awards its rightful place and profile on
      television," Adam Perry, Channel 5's controller of events, said in a statement.</p>
    <p>Initial, the independent production company that will produce coverage of the awards,
      said it aims to bring in top celebrities and world-class performers for the event.</p>
    <p>The British Fashion Awards will be held on October 22 at London's Royal Albert Hall.</p>
    <p>-- London Advertising Newsdesk +44 171 542 2815</p>
  </text>
  <copyright>(c) Reuters Limited 1997</copyright>
</newsitem>
```

The news document itemid: 783802 contains: 122 words

Channel 5, Britain's newest terrestrial television channel, on Wednesday said it struck a three-year deal with the British Fashion Council for exclusive rights to broadcast the Lloyds Bank British Fashion Awards.

"Fashion is now a defining part of British popular culture and this was an unmissable opportunity for 5 to give the British Fashion Awards its rightful place and profile on television," Adam Perry, Channel 5's controller of events, said in a statement.

Initial, the independent production company that will produce coverage of the awards, said it aims to bring in top celebrities and world-class performers for the event.

The British Fashion Awards will be held on October 22 at London's Royal Albert Hall.

-- London Advertising Newsdesk +44 171 542 2815

Select the sentence in the XML document

- The XML document
 - “783802.xml”
- The sentence
 - “*The British Fashion Awards will be held on October 22 at London's Royal Albert Hall.*”

Now we define a function to get or select the sentence.

PS: The strip() method removes any leading, and trailing whitespaces.

```
def select_sentence(inputpath, stop_ws):
    myfile=open('783802.xml')
    selected_s=""
    start_end = False
    file_=myfile.readlines()
    for line in file_:
        line = line.strip()
        if(start_end == False):
            if line.startswith("<text>"):
                start_end = True
        elif line.startswith("</text>"):
            break
        else:
            if line.startswith("<p>The"):
                selected_s = line
    myfile.close()
    return(selected_s)
```

Parse the sentence and represent it using different data structures (long string and list of strings)

Now we go back to the example for the tokenization process. This time, we get the line (sentence) by calling the defined function.

```
import sys
stopwords_f = open('common-english-words.txt', 'r')
stop_words = stopwords_f.read().split(',')
stopwords_f.close()
line = select_sentence("", stop_words)
line = line.replace("<p>", "").replace("</p>", "")
line = line.translate(str.maketrans('', '', string.digits)).translate(str.maketrans(string.punctuation, ' '*len(string.punctuation)))
print(line)
words=line.split()
print(words)
terms=[word.lower() for word in words if len(word)>2]
print(terms)
```

The British Fashion Awards will be held on October at London s Royal Albert Hall

['The', 'British', 'Fashion', 'Awards', 'will', 'be', 'held', 'on', 'October', 'at', 'London', 's', 'Royal', 'Albert', 'Hall']

['the', 'british', 'fashion', 'awards', 'will', 'held', 'october', 'london', 'royal', 'albert', 'hall']

7. Teaching Methodology

Teacher-Centered Learning vs. Student-Centered Learning

- Lecturers choose **what** students will learn (e.g., curriculum design), and **how** the students will be assessed (e.g., assessment items and making guides).
- **How will students learn?**
 - Develop learner autonomy and independence
 - focus on skills and practices that enable independent ICT problem-solving (Search-based solutions and AI-based solutions)
 - Assess your own learning by reviewing lecture notes, answering review questions and workshop tasks, and
 - to be active, responsible participation in workshops
 - with your own pace of learning - you may choose on-campus workshop or on-line workshop, attend real time lectures and/or listen to lecture recordings; and
 - Complete all assessment items.

About Workshops

- Lab based workshops.
- Practical exercise tasks are on the CANVAS, workshop will start from week 2.
- You may be required to research some commands or algorithms supervised by your tutor in workshop.
- In this week, I strongly recommend you read lecture notes (PART 2) carefully **if you are not very confident about using Python**.
- I also strongly recommend you complete all practical exercises (tasks) and review questions.
- If you have problems, please let me or your tutor know (All contact information can be found in the IFN647 Canvas) by using emails or IFN647 Slack.
- We do not use a fixed textbook. Instead, key references are provided each week, for example,
 - Traditional topics, e.g., W. Bruce Croft et al., Search Engines: Information Retrieval in Practice, Pearson, 2010.
 - Recent developments, e.g., widely cited and influential references on large language models (LLMs).

Policies of Using GenAI for Assessment Items

- Assignment 1 (Individual, 20%)
 - The assignment specification outlines restrictions on the use of Python packages and specifies required data structures.
 - The use of GenAI tools is prohibited during this assessment.
- Assignment 2 (Group, 35%).
 - It uses a specialized data collection consisting of multiple datasets, queries, and corresponding relevance judgements. Students are required to develop three models: two baseline models and one new model. The baseline models are clearly specified, while there are no explicit constraints on how the new model should be developed.
 - The objective of the new model is to outperform the baseline models according to selected effectiveness measures and significant test (evaluation).
 - The ethical and responsible use of GenAI tools is authorized in this assessment.
- Final Exam (45%): A written examination.
 - The use of GenAI tools is prohibited during this assessment.

References

- [1] [Dan Hendrycks](#), Introduction to AI Safety, Ethics, and Society, 2025 (2nd version), <https://doi.org/10.1201/9781003530336>
- [2] W. Bruce Croft, *Search Engines - Information retrieval in Practice*; Pearson, 2010.
https://qut.primo.exlibrisgroup.com/permalink/61QUT_INST/1tofoc8/alma991002142279704001 or
<http://ciir.cs.umass.edu/downloads/SEIRiP.pdf>
- [3] Wenqi Fan, Yujuan Ding, Liangbo Ning, Shijie Wang, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. 2024. A Survey on RAG Meeting LLMs, 6491–6501. <https://doi.org/10.1145/3637528.3671470>
- [4] Larisa Gorenstein, Eli Konen, Michael Green, Eyal Klang, *Bidirectional Encoder Representations from Transformers in Radiology: A Systematic Review of Natural Language Processing Applications*, Journal of the American College of Radiology, Volume 21, Issue 6, 2024, Pages 914-941, <https://doi.org/10.1016/j.jacr.2024.01.012>
- [5] Vasant Dhar, Data Science and Prediction, *Communications of the ACM*, Vol. 56 No. 12, Pages 64-73, 2013, <https://dl.acm.org/doi/10.1145/2500499>
- [6] Pedro Domingos, A Few Useful Things to Know About Machine Learning, *Communications of the ACM*, 2012, vol. 55, no. 10, pages 78-87. <https://dl.acm.org/doi/10.1145/2347736.2347755>
- [7] Thiebes, S., Lins, S. & Sunyaev, A. Trustworthy artificial intelligence. *Electron Markets* **31**, 447–464 (2021). <https://doi.org/10.1007/s12525-020-00441-4>
- [8] Lo Piano, S. Ethical principles in machine learning and artificial intelligence: cases from the field and possible ways forward. *Humanit Soc Sci Commun* **7**, 9 (2020). <https://doi.org/10.1057/s41599-020-0501-9>